

Student Performance

Correlation & Causation Study

500-Student Synthetic Dataset | Pearson Correlation Analysis

500

Students

4

Variables

2026

Year

1. EXECUTIVE SUMMARY

This report analyzes student data across four dimensions — Study Hours, Sleep Hours, Attendance, and Past Grades — to understand their relationship with current GPA and Placement Status. The goal is to identify which factors most strongly correlate with academic success.

2. METHODOLOGY

Data Collection	Synthetic dataset representing 500 students.
Data Cleaning	Missing values in Study Hours and Sleep Hours imputed via median. Outliers clipped to realistic maxima.
Statistical Method	Pearson correlation used to measure linear relationships between variables.

3. KEY FINDINGS

Study Hours vs GPA Strong positive correlation. Students who study more consistently achieve higher GPAs with a direct linear relationship. $r \sim 0.90$	Sleep Hours vs GPA Moderate positive correlation. Adequate sleep contributes positively, though less dramatically than study time. $r \sim 0.40$
Attendance vs GPA Weak to moderate correlation. Regular attendance supports learning but doesn't guarantee high grades alone. $r \sim 0.35$	Past Grades vs GPA Very strong correlation. Past performance is the most reliable predictor, reflecting stable student habits. $r \sim 0.95$

4. RECOMMENDATIONS

1	■ Focus on Study Time Institutions should provide better environments — libraries, study groups — to encourage longer, more effective study sessions.
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2	■ Promote Healthy Sleep Awareness campaigns about the importance of 7–8 hours of sleep could yield noticeable improvements in student grades.
3	■ Early Intervention Since Past Grades strongly predict current GPA, students with historically lower grades should be targeted for early tutoring and intervention programs.

5. LIMITATIONS

*This study relies on **synthetic data**. Real-world relationships often involve confounding variables — student stress, socio-economic background, course difficulty — not captured here. Furthermore, **correlation does not imply causation**: for instance, a high GPA might motivate better attendance, rather than attendance directly causing higher grades.*

Variable	Correlation (r)	Strength	Impact
Past Grades	~0.95	Very Strong	★★★★★
Study Hours	~0.90	Strong	★★★★★
Sleep Hours	~0.40	Moderate	★★★██
Attendance	~0.35	Weak–Moderate	★★★██